Question: 16 TO IMPLEMENT A PRIORITY QUEUE USING AN ARRAY IN C, WHERE ELEMENTS ARE DELETED BASED ON THE HEIGHEST PRIORITY

main.c   Share   Run

```c
#include <stdio.h>
#define MAX 100
int size = 0;
int pq[MAX];
int prio[MAX];
void insert(int value, int priority) {
    pq[size] = value;
    prio[size] = priority;
    size++;
}
int getHighestPriorityIndex() {
    int maxIndex = 0;
    for (int i = 1; i < size; i++) {
        if (prio[i] > prio[maxIndex]) {
            maxIndex = i;
        }
    }
    return maxIndex;
}
int deleteHighestPriority() {
    int index = getHighestPriorityIndex();
    int value = pq[index];
    for (int i = index; i < size - 1; i++) {
        pq[i] = pq[i + 1];
        prio[i] = prio[i + 1];
    }
```

Output

```
10(2) 20(5) 30(1) 40(4)
20
40
10(2) 30(1)


=== Code Execution Successful ===
```

main.c   Share   Run

```c
    int index = getHighestPriorityIndex();
    int value = pq[index];
    for (int i = index; i < size - 1; i++) {
        pq[i] = pq[i + 1];
        prio[i] = prio[i + 1];
    }
    size--;
    return value;
}
void display() {
    for (int i = 0; i < size; i++) {
        printf("%d(%d) ", pq[i], prio[i]);
    }
    printf("\n");
}
int main() {
    insert(10, 2);
    insert(20, 5);
    insert(30, 1);
    insert(40, 4);
    display();
    printf("%d\n", deleteHighestPriority());
    printf("%d\n", deleteHighestPriority());
    display();
    return 0;
}
```